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# Direct Self-evolving Optimization: Efficient Self-Evolving LLM Training without Challenger Model

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## Abstract

1 The self-evolving training paradigm has emerged as a crucial technique for en-  
2 hancing the ability of Large Language Models (LLMs) to perform mathematical  
3 reasoning and handle complex tasks. By training a challenger model to generate  
4 questions for a solver model, this approach effectively overcomes the scarcity of  
5 human-annotated data. However, a major drawback of this paradigm is the signifi-  
6 cant increase in computational and memory costs. In this paper, we eliminate the  
7 need for challenger model training by deriving a closed-form optimal challenger  
8 policy, conditioned on the current solver model. We show that this optimal policy  
9 is a reweighted version of the base model’s policy, where the weights correspond  
10 to the challenger’s reward. Finally, we employ the Metropolis-Hastings algorithm  
11 to iteratively sample questions from this optimal policy, which are subsequently  
12 fed into the solver model for efficient training.

## 13 1 Introduction

### 14 1.1 Existing method

15 In the existing self-evolving LLM training [Huang et al., 2025, Xia et al., 2025], the **challenger**  
16 optimizes:

$$\max_{\phi} R_c(\phi, \theta) := \mathbb{E}_{x \sim \pi_{\phi}(\cdot | p_0)} [r_{c,m}(x, \theta)] - \beta \text{KL}(\pi_{\phi}(\cdot | p_0) \parallel \pi_{\text{base}}(\cdot | p_0)) \quad (1)$$

17 where  $r_{c,m}(x, \theta)$  is the expected challenger reward obtained from  $m$  stochastic solver responses to a  
18 fixed question  $x$ . In particular,

$$r_{c,m}(x, \theta) := \mathbb{E}_{\xi \sim P_{\theta}^{\otimes m}(\cdot | x)} [\hat{r}_{c,m}(x, \theta; \xi)], \quad (2)$$

19 where  $\hat{r}_{c,m}$  is the empirical uncertainty reward defined in equation 11. Thus, the expectation over  
20  $x$  is the outer expectation in equation 1, whereas the expectation over the rollout randomness  $\xi$  is  
21 conditional on a fixed question.

22 The **solver** optimizes:

$$\max_{\theta} R_s(\phi, \theta) := \mathbb{E}_{x \sim \pi_{\phi}} [\mathbb{E}_{y \sim \pi_{\theta}(y | x)} [r_s(x, y)]] \quad (3)$$

23 where  $r_s(x, y)$  is the solver’s reward function.

24 Since the two objectives depend on both  $\phi$  and  $\theta$ , existing solution is to fix one and update another.  
25 Given initialization  $\phi_0$  and  $\theta_0$ , the updating rule is

$$\begin{aligned} \phi_{t+1} &\leftarrow \text{update } \phi_t \text{ for K steps using GRPO on reward } R_c(\phi_t, \theta_t), \\ \theta_{t+1} &\leftarrow \text{update } \theta_t \text{ for K steps using GRPO on reward } R_s(\phi_{t+1}, \theta_t) \end{aligned}$$

## 26 1.2 The Weakness and Our proposal

27 However, training two models are costly and this alternating optimization is unstable. To optimize  
 28 equation 1 and equation 3, we first derive the optimal challenger distribution for a fixed solver  $\pi_\theta$ .  
 29 Write  $q(x) := \pi_\phi(x \mid p_0)$  and  $\pi_0(x) := \pi_{\text{base}}(x \mid p_0)$ . Relaxing the challenger parameterization,  
 30 equation 1 becomes

$$\max_{q \in \mathcal{P}(\text{supp } \pi_0)} J_\theta(q) := \sum_x q(x) r_{c,m}(x, \theta) - \beta \sum_x q(x) \log \frac{q(x)}{\pi_0(x)}, \quad \beta > 0, \quad (4)$$

31 where  $\mathcal{P}(\text{supp } \pi_0)$  is the set of probability distributions supported on  $\text{supp } \pi_0$ . Introducing a Lagrange  
 32 multiplier  $\gamma$  for  $\sum_x q(x) = 1$  gives

$$\mathcal{L}(q, \gamma) = \sum_x q(x) r_{c,m}(x, \theta) - \beta \sum_x q(x) \log \frac{q(x)}{\pi_0(x)} + \gamma \left( \sum_x q(x) - 1 \right). \quad (5)$$

33 For every  $x$  in the support of  $\pi_0$ , stationarity with respect to  $q(x)$  yields

$$0 = \frac{\partial \mathcal{L}}{\partial q(x)} = r_{c,m}(x, \theta) - \beta \left( \log \frac{q(x)}{\pi_0(x)} + 1 \right) + \gamma. \quad (6)$$

34 Consequently,

$$q(x) = \pi_0(x) \exp \left( \frac{r_{c,m}(x, \theta)}{\beta} + \frac{\gamma}{\beta} - 1 \right).$$

35 Enforcing normalization determines the constant and gives

$$q_{\beta, \theta, m}^*(x) = \frac{\pi_{\text{base}}(x \mid p_0) \exp(r_{c,m}(x, \theta)/\beta)}{Z_m(\theta, \beta)}, \quad Z_m(\theta, \beta) := \mathbb{E}_{x \sim \pi_{\text{base}}(\cdot \mid p_0)} [\exp(r_{c,m}(x, \theta)/\beta)]. \quad (7)$$

36 This stationary point is the global optimum. Indeed, substituting equation 7 into equation 4 gives, for  
 37 any feasible  $q$ ,

$$J_\theta(q) = \beta \log Z_m(\theta, \beta) - \beta \text{KL}(q \parallel q_{\beta, \theta, m}^*) \leq \beta \log Z_m(\theta, \beta), \quad (8)$$

38 with equality if and only if  $q = q_{\beta, \theta, m}^*$ . Therefore, if the challenger policy class can represent  $q_{\beta, \theta, m}^*$ ,  
 39 then

$$\phi^*(\theta) \in \arg \max_{\phi} R_c(\phi, \theta) \quad \text{and} \quad \pi_{\phi^*}(\theta)(x \mid p_0) = q_{\beta, \theta, m}^*(x).$$

40 We then plug  $q_{\beta, \theta, m}^*$  into equation 3:

$$\begin{aligned} \max_{\theta} R(\theta) &:= R_s(\phi^*(\theta), \theta) \\ &= \mathbb{E}_{x \sim q_{\beta, \theta, m}^*} [\mathbb{E}_{y \sim \pi_\theta(\cdot \mid x)} [r_s(x, y)]] \\ &= \frac{1}{Z_m(\theta, \beta)} \mathbb{E}_{x \sim \pi_{\text{base}}(\cdot \mid p_0)} \left[ \exp \left( \frac{r_{c,m}(x, \theta)}{\beta} \right) \mathbb{E}_{y \sim \pi_\theta(\cdot \mid x)} [r_s(x, y)] \right]. \end{aligned} \quad (9)$$

41 To distinguish the finite-rollout objective from an idealized population uncertainty, let  $g(y)$  extract  
 42 and canonicalize the final answer from a solver response  $y$ , and define

$$\begin{aligned} P_\theta(a \mid x) &:= \Pr_{y \sim \pi_\theta(\cdot \mid x)} \{g(y) = a\}, \\ p_\theta^*(x) &:= \max_a P_\theta(a \mid x), \\ r_c^{\text{pop}}(x, \theta) &:= 1 - 2 \left| p_\theta^*(x) - \frac{1}{2} \right|. \end{aligned} \quad (10)$$

43 For  $m$  i.i.d. responses  $S_\theta(x; \xi) = \{y_1, \dots, y_m\}$ , define the empirical modal probability and uncer-  
 44 tainty reward by

$$\begin{aligned}\hat{p}_{\theta,m}(x; \xi) &:= \max_a \frac{1}{m} \sum_{j=1}^m \mathbb{I}\{g(y_j) = a\}, \\ \hat{r}_{c,m}(x, \theta; \xi) &:= 1 - 2 \left| \hat{p}_{\theta,m}(x; \xi) - \frac{1}{2} \right|, \quad r_{c,m}(x, \theta) := \mathbb{E}_\xi[\hat{r}_{c,m}(x, \theta; \xi)].\end{aligned}\quad (11)$$

45 Because the maximum and absolute value are nonlinear, generally

$$r_{c,m}(x, \theta) \neq r_c^{\text{pop}}(x, \theta) \quad \text{for finite } m, \quad r_{c,m}(x, \theta) \longrightarrow r_c^{\text{pop}}(x, \theta) \quad \text{as } m \rightarrow \infty. \quad (12)$$

46 The stochastic Challenger update used in R-Zero optimizes  $r_{c,m}$  in expectation;  $r_c^{\text{pop}}$  is its idealized  
 47 large- $m$  limit.

48 We handle diversity at the batch level. For a batch  $X = (x_1, \dots, x_N)$ , compute  $d_{ij} = 1 -$   
 49 BLEU( $x_i, x_j$ ) and cluster questions for which  $d_{ij} < \tau_{\text{BLEU}}$ . Let  $C_X(x_i)$  denote the cluster contain-  
 50 ing  $x_i$  and define

$$r_{\text{rep}}(x_i; X) := \frac{|C_X(x_i)|}{N}. \quad (13)$$

51 The coefficient of this penalty is denoted by  $\lambda_{\text{rep}} \geq 0$  below; this notation distinguishes it from the  
 52 dual variable used in the adaptive-temperature update.

53 For  $r_s(x, y_j)$ , it is 1 if  $g(y_j)$  is identical to the modal pseudo-label extracted from the  $m$  responses,  
 54 and 0 otherwise.

55 Hence, we can directly optimize this objective using policy gradient. First, we generate a mini-batch  
 56  $X = (x_1, \dots, x_N)$  from the base model  $\pi_{\text{base}}(\cdot | p_0)$ . For each question, the solver model generates  
 57  $m$  responses  $\{y_{i,1}, \dots, y_{i,m}\}$ . Treating the sampling weights as fixed during the solver update, we  
 58 estimate the policy gradient as

$$\hat{\nabla}_\theta R(\theta) = \frac{1}{N} \sum_{i=1}^N \exp\left(\frac{r_{c,m}(x_i, \theta)}{\beta}\right) \frac{1}{m} \sum_{j=1}^m r_s(x_i, y_{i,j}) \nabla_\theta \log \pi_\theta(y_{i,j} | x_i). \quad (14)$$

59 We directly optimize the solver model. It is a weighted policy gradient method, where the weights  
 60 depend on how the current solver model performs on the generated questions. For the questions that  
 61 the solver has higher uncertainty, its gradient will be up-weighted.

### 62 1.3 Active Sampling with Noisy Rewards

63 Equation 7 suggests sampling questions from the exponentially tilted base distribution.  
 64 Because the repetition penalty in equation 13 couples questions within a batch, we perform MCMC  
 65 directly on the batch state  $X = (x_1, \dots, x_N) \in \mathcal{X}^N$ . Let

$$\begin{aligned}\Pi_{\text{base}}(X | p_0) &:= \prod_{i=1}^N \pi_{\text{base}}(x_i | p_0), \\ \hat{u}_{\theta,m}(x_i; X, \xi_i) &:= [\hat{r}_{c,m}(x_i, \theta; \xi_i) - \lambda_{\text{rep}} r_{\text{rep}}(x_i; X)]_+, \\ \hat{U}_{\theta,m}(X; \xi) &:= \sum_{i=1}^N \hat{u}_{\theta,m}(x_i; X, \xi_i), \quad U_{\theta,m}(X) := \mathbb{E}_\xi[\hat{U}_{\theta,m}(X; \xi)].\end{aligned}\quad (15)$$

66 Here  $\xi = (\xi_1, \dots, \xi_N)$  collects all solver-rollout randomness, and  $[a]_+ := \max\{a, 0\}$ . The batch-  
 67 level target is

$$q_{\beta,\theta,m}(X) = \frac{1}{Z_{N,m}(\theta, \beta)} \Pi_{\text{base}}(X | p_0) \exp\left(\frac{U_{\theta,m}(X)}{\beta}\right). \quad (16)$$

68 Let  $K(X' | X)$  be the batch mutation kernel. The exact Metropolis–Hastings log ratio is

$$A_{\beta, \theta, m}(X, X') := \ell(X, X') + \frac{U_{\theta, m}(X') - U_{\theta, m}(X)}{\beta},$$

$$\ell(X, X') := \log \frac{\Pi_{\text{base}}(X' | p_0) K(X | X')}{\Pi_{\text{base}}(X | p_0) K(X' | X)}. \quad (17)$$

69 Thus,  $\alpha(X, X') = \min\{1, \exp(A_{\beta, \theta, m}(X, X'))\}$ . In our implementation, we use the approximation  
 70  $\ell(X, X') \approx 0$ , corresponding to treating the mutation kernel as approximately reversible with respect  
 71 to  $\Pi_{\text{base}}$ . If only symmetry of  $K$  is assumed, the base-density ratio in equation 17 must still be  
 72 retained.

73 **Why naive noisy MH is biased.** The utility  $U_{\theta, m}(X)$  is unavailable because it is an expectation  
 74 over solver rollouts. Substituting one estimate directly into the acceptance ratio gives

$$\hat{\alpha}_{\text{naive}}(X, X'; \xi, \xi') = \min \left\{ 1, \exp \left( \ell(X, X') + \frac{\hat{U}_{\theta, m}(X'; \xi') - \hat{U}_{\theta, m}(X; \xi)}{\beta} \right) \right\}. \quad (18)$$

75 In general, this does not target equation 16. In particular, exponentiating before averaging yields

$$\mathbb{E}[\exp(\hat{U}_{\theta, m}/\beta)] \quad \text{rather than} \quad \exp(\mathbb{E}[\hat{U}_{\theta, m}]/\beta),$$

76 which favors states with high reward variance.

## 77 1.4 Ceperley–Dewing Penalty Acceptance

78 We use the penalty method of Ceperley and Dewing [1999] to correct the leading bias caused by  
 79 noisy utilities. Write the exact utility difference as

$$\Delta_{\theta, m}(X, X') := U_{\theta, m}(X') - U_{\theta, m}(X). \quad (19)$$

80 Suppose that an unbiased estimator  $\tilde{D}$  of  $\Delta_{\theta, m}(X, X')$  is approximately Gaussian with variance  $\sigma_D^2$ .  
 81 If  $\sigma_D^2$  were known, the penalty acceptance probability would be

$$\hat{\alpha}_{\text{CD}}(X, X') = \min \left\{ 1, \exp \left( \ell(X, X') + \frac{\tilde{D}}{\beta} - \frac{\sigma_D^2}{2\beta^2} \right) \right\}. \quad (20)$$

82 Indeed, if

$$\tilde{D} | X, X' \sim \mathcal{N}(\Delta_{\theta, m}(X, X'), \sigma_D^2), \quad (21)$$

83 then

$$\mathbb{E} \left[ \exp \left( \frac{\tilde{D}}{\beta} - \frac{\sigma_D^2}{2\beta^2} \right) \middle| X, X' \right] = \exp \left( \frac{\Delta_{\theta, m}(X, X')}{\beta} \right). \quad (22)$$

84 Under the Gaussian-noise assumption, the resulting penalty method satisfies the required noisy  
 85 detailed-balance identity.

### 86 1.4.1 A rollout-efficient U-statistic estimator

87 Repeating an entire set of  $m$  solver rollouts many times is expensive. Instead, for every question  $x_i$   
 88 that must be evaluated, generate only  $n = m + k$  i.i.d. responses  $Y_i^{(n)} = (y_{i,1}, \dots, y_{i,n})$ , where  $k$  is  
 89 small. Let  $\mathcal{S}_{m,n} := \{S \subseteq [n] : |S| = m\}$  and define the order- $m$  U-statistic [Hoeffding, 1948]

$$\tilde{u}_{\theta, m, n}(x_i; X, Y_i^{(n)}) := \binom{n}{m}^{-1} \sum_{S \in \mathcal{S}_{m,n}} [\hat{r}_{c, m}(x_i, \theta; Y_{i,S}) - \lambda_{\text{rep}} r_{\text{rep}}(x_i; X)]_+,$$

$$\tilde{U}_{\theta, m, n}(X) := \sum_{i=1}^N \tilde{u}_{\theta, m, n}(x_i; X, Y_i^{(n)}). \quad (23)$$

90 The positive part and repetition penalty are included inside the U-statistic kernel because they are  
 91 part of the batch utility in equation 15. Averaging the uncertainty score first and applying the positive  
 92 part afterward would in general estimate a different objective. By symmetry of the i.i.d. responses,

$$\mathbb{E}[\tilde{U}_{\theta,m,n}(X) \mid X] = U_{\theta,m}(X). \quad (24)$$

93 Thus, increasing  $n$  reduces variance without changing the finite- $m$  target  $r_{c,m}$ . It does not remove  
 94 the finite- $m$  discrepancy in equation 12.

95 For independently evaluated current and proposed states, set

$$\tilde{D}_{m,n}(X, X') := \tilde{U}_{\theta,m,n}(X') - \tilde{U}_{\theta,m,n}(X). \quad (25)$$

96 To estimate its variance without additional solver calls, use the delete-one jackknife. For  $n > m$ , let  
 97  $\tilde{u}_i^{(-j)}(X)$  be the same U-statistic as in equation 23, computed after deleting response  $j$ , and define

$$\begin{aligned} \bar{u}_i^{(-)}(X) &:= \frac{1}{n} \sum_{j=1}^n \tilde{u}_i^{(-j)}(X), \\ \hat{v}_i(X) &:= \frac{n-1}{n} \sum_{j=1}^n \left( \tilde{u}_i^{(-j)}(X) - \bar{u}_i^{(-)}(X) \right)^2, \\ \hat{\sigma}_X^2 &:= \sum_{i=1}^N \hat{v}_i(X), \\ \hat{\sigma}_D^2 &:= \hat{\sigma}_{X'}^2 + \hat{\sigma}_X^2. \end{aligned} \quad (26)$$

98 The last two identities use independent solver samples across questions and independent evaluations  
 99 of  $X$  and  $X'$ . With paired evaluations, the corresponding jackknife covariance term is subtracted  
 100 from  $\hat{\sigma}_D^2$ .

101 Our practical plug-in CD rule is therefore

$$\hat{\alpha}_{\text{U-CD}}(X, X') = \min \left\{ 1, \exp \left( \ell(X, X') + \frac{\tilde{D}_{m,n}(X, X')}{\beta} - \frac{\hat{\sigma}_D^2}{2\beta^2} \right) \right\}. \quad (27)$$

102 Because equation 27 replaces the true variance by a jackknife estimate and relies on approximate  
 103 Gaussianity, it defines an approximate rather than exact MH kernel. We increase  $n$  adaptively until  
 104  $\hat{\sigma}_D/\beta \leq \varepsilon_{\text{noise}}$ , subject to a maximum rollout budget  $n_{\text{max}}$ . For example, when  $m = 9$ ,  $n = 12$   
 105 requires only three additional responses per evaluated question and averages  $\binom{12}{9} = 220$  correlated  
 106 subsets; the jackknife, not the raw number of subsets, determines the variance estimate. When  $\binom{n}{m}$   
 107 is too large, an incomplete U-statistic can average a uniform random sample of subsets, with its  
 108 additional Monte Carlo variance included in  $\hat{\sigma}_D^2$ .

109 For a coordinate-wise proposal that replaces only one question, most batch contributions cancel. No  
 110 solver calls are needed for questions whose question text and repetition penalty are unchanged. For  
 111 unchanged questions whose repetition penalty changes, one fresh response set can be shared between  
 112 the current and proposed utilities as a paired evaluation; in this case we jackknife the full paired  
 113 difference so that its covariance is retained. Fresh response sets are used at each MCMC step in the  
 114 penalty-method version; caching a previously accepted noisy utility would instead define a different  
 115 approximate auxiliary-variable chain and may create sticky high-noise states.

116 **Zero-additional-rollout approximation.** If even  $k > 0$  is unaffordable, retain the original  $m$   
 117 responses and bootstrap them within each question. Recompute the entire batch difference, including  
 118 the positive part and repetition penalty, for  $B_{\text{boot}}$  bootstrap resamples and use their sample variance  
 119 as  $\hat{\sigma}_{D,\text{boot}}^2$  in equation 27, while replacing  $\tilde{D}_{m,n}$  by the original single-evaluation difference. This  
 120 requires no additional LLM calls, but it estimates rather than reduces reward noise and should not be  
 121 interpreted as  $B_{\text{boot}}$  independent evaluations. We therefore regard the bootstrap rule as a cheaper,  
 122 more strongly approximate ablation.

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**Algorithm 1: U-statistic Penalty MCMC on Batch States**


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**Input:** initial batch  $X^0$ , solver  $\pi_\theta$ , mutation kernel  $K$ , temperature  $\beta$ , MCMC length  $\tau$ , rollout counts  $m < n \leq n_{\max}$ , noise tolerance  $\varepsilon_{\text{noise}}$

**for**  $s = 0, \dots, \tau - 1$  **do**

    Sample a proposed batch  $X' \sim K(\cdot \mid X^s)$

    Generate fresh solver responses for all stochastic utility terms needed to compare  $X^s$  and  $X'$

    Compute  $\tilde{U}_{\theta,m,n}(X^s)$  and  $\tilde{U}_{\theta,m,n}(X')$  using equation 23

    Compute  $\tilde{D}_{m,n}(X^s, X')$  and  $\hat{\sigma}_D^2$  using equation 25–equation 26

**while**  $\hat{\sigma}_D/\beta > \varepsilon_{\text{noise}}$  **and**  $n < n_{\max}$  **do**

        Generate additional responses and recompute the U-statistics and jackknife variance

**end**

    Compute  $\ell(X^s, X')$  using equation 17, or set it to zero under the base-reversibility approximation

    Set  $a = \min\{0, \ell(X^s, X') + \tilde{D}_{m,n}(X^s, X')/\beta - \hat{\sigma}_D^2/(2\beta^2)\}$

    Set  $X^{s+1} = X'$  with probability  $\exp(a)$ ; otherwise set  $X^{s+1} = X^s$

**end**

**Output:**  $X^\tau$

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**Algorithm 2: MCMC-DEO**


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**Input:** initial solver  $\theta_0$ , question-generating prompt  $p_0$ , learning rate  $\eta$ , temperature  $\beta$

**for**  $t = 0, \dots, T - 1$  **do**

    Sample  $X_t^0 = (x_{t,1}^0, \dots, x_{t,N}^0) \sim \Pi_{\text{base}}(\cdot \mid p_0)$

    Obtain  $X_t^\tau$  by Algorithm 1 with solver  $\pi_{\theta_t}$

    Generate fresh solver responses for each question in  $X_t^\tau$  and construct the solver-training gradient  $\hat{\nabla}_\theta R(\theta_t)$

    Update  $\theta_{t+1} = \theta_t + \eta \hat{\nabla}_\theta R(\theta_t)$

**end**

**Output:**  $\theta_T$

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123 **1.5 Adaptive Temperature Update**

124 We control question hardness by requiring most uncertainty scores to lie in  $[r_{\min}, r_{\max}]$ . Define the  
125 in-band fraction

$$H_{\theta,m}(X) := \frac{1}{N} \sum_{i=1}^N \mathbb{I}\{r_{\min} \leq r_{c,m}(x_i, \theta) \leq r_{\max}\}. \quad (28)$$

126 For a prescribed violation rate  $\delta$ , we choose the largest admissible temperature:

$$\max_{\beta \in [\beta_{\min}, \beta_{\max}]} \beta \quad \text{subject to} \quad 1 - \mathbb{E}_{X \sim q_{\beta,\theta,m}}[H_{\theta,m}(X)] \leq \delta. \quad (29)$$

127 Equivalently, introduce a dual variable  $\nu \geq 0$  and solve

$$\min_{\beta \in [\beta_{\min}, \beta_{\max}]} \max_{\nu \geq 0} \mathcal{L}(\beta, \nu) := -\log \beta + \nu (1 - \mathbb{E}_{q_{\beta,\theta,m}}[H_{\theta,m}(X)] - \delta). \quad (30)$$

128 By equation 16,

$$\frac{\partial}{\partial \beta} \log q_{\beta,\theta,m}(X) = -\frac{1}{\beta^2} (U_{\theta,m}(X) - \mathbb{E}_{q_{\beta,\theta,m}}[U_{\theta,m}(X)]). \quad (31)$$

129 Therefore,

$$\nabla_\beta \mathcal{L}(\beta, \nu) = -\frac{1}{\beta} + \frac{\nu}{\beta^2} \text{Cov}_{q_{\beta,\theta,m}}(H_{\theta,m}(X), U_{\theta,m}(X)), \quad (32)$$

$$\nabla_\nu \mathcal{L}(\beta, \nu) = 1 - \mathbb{E}_{q_{\beta,\theta,m}}[H_{\theta,m}(X)] - \delta. \quad (33)$$

130 Given  $B$  MCMC batch states  $X^{(1)}, \dots, X^{(B)} \sim q_{\beta,\theta,m}$ , estimate their utilities by

$$\tilde{U}_b := \tilde{U}_{\theta,m,n}(X^{(b)}), \quad b = 1, \dots, B,$$

131 and let

$$\overline{H}_B := \frac{1}{B} \sum_{b=1}^B H_{\theta,m}(X^{(b)}), \quad \overline{U}_B := \frac{1}{B} \sum_{b=1}^B \tilde{U}_b.$$

132 The empirical gradients are

$$\widehat{\nabla}_\beta \mathcal{L} = -\frac{1}{\beta} + \frac{\nu}{\beta^2(B-1)} \sum_{b=1}^B \left( H_{\theta,m}(X^{(b)}) - \overline{H}_B \right) \left( \tilde{U}_b - \overline{U}_B \right), \quad (34)$$

$$\widehat{\nabla}_\nu \mathcal{L} = 1 - \overline{H}_B - \delta. \quad (35)$$

133 Using an inner-loop index  $k$  to distinguish temperature adaptation from the outer solver-training  
134 index  $t$ , the descent-ascent updates are

$$\begin{aligned} \beta^{k+1} &= \mathcal{P}_{[\beta_{\min}, \beta_{\max}]} \left( \beta^k - \eta_\beta \widehat{\nabla}_\beta \mathcal{L}(\beta^k, \nu^k) \right), \\ \nu^{k+1} &= \left[ \nu^k + \eta_\nu \widehat{\nabla}_\nu \mathcal{L}(\beta^k, \nu^k) \right]_+. \end{aligned} \quad (36)$$

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